# ECEN 607: Advanced Analog Circuit Tech

# Lab5: Two-stage Folded-Cascode OPAMP With Class AB output stage

## Lab-Report

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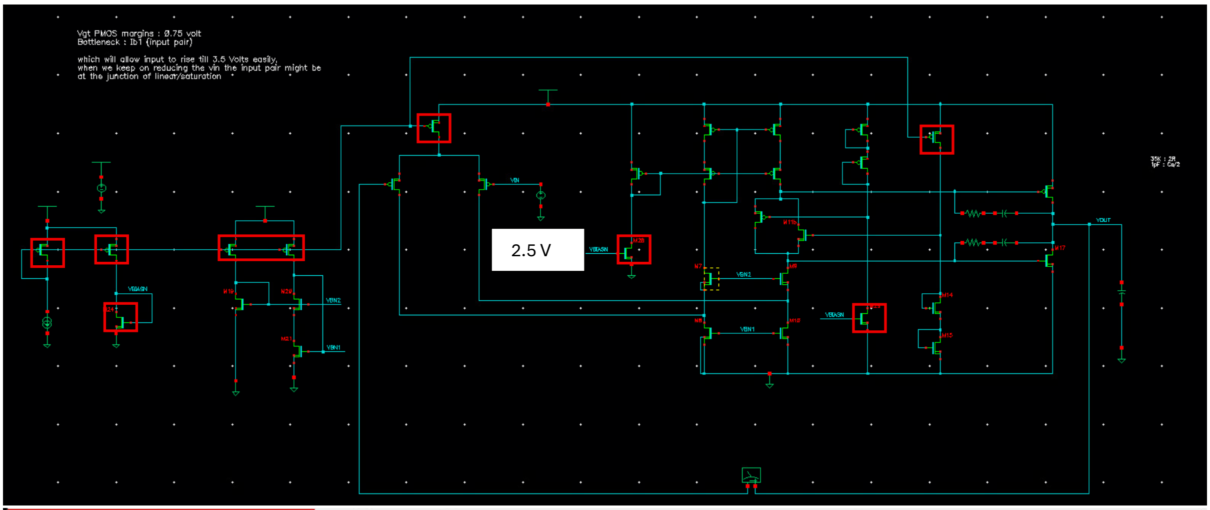
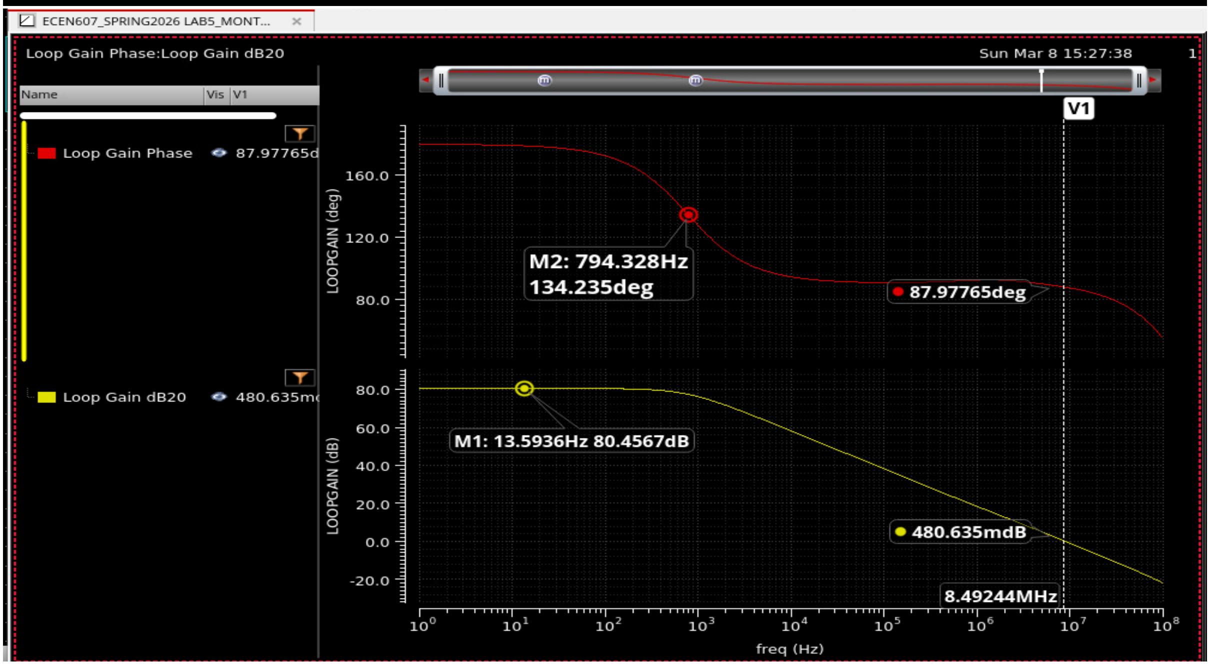
**Section: 602**

1. **Prelab-design simulation:**

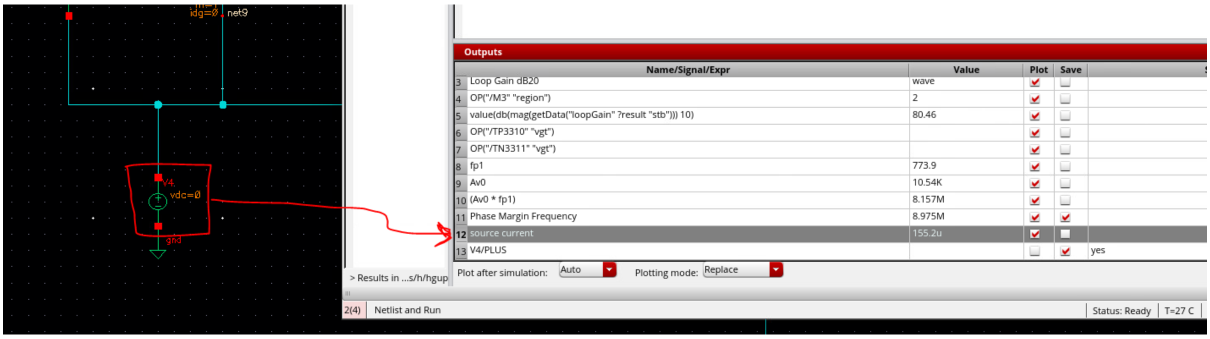
   **<u>Design changes done upfront</u>**: There were 6 ideal current sources (Ib1, Ib2, Ib3, Ib4, Ib5, Ib6) shown in my-prelab. Here it was modified such that there is only one ideal current-source and everywhere else this was used to get current by the means of current mirrors. (**all the highlighted boxes below highlight the transistor added to emulate the bias current sources**), they were sized for good matching (higher Vdsat, longer length, lower gm). There sizes will be present in the Part-3 of this manual.

   **<u>Test-bench (Used):</u>** Just for checking the bias-margins, gain, phase margins, power consumption. 2.5 Volts input voltage was applied, CL = 20pF.




M1: DC gain 80.4567 dB (> SPEC (60 dB), ~ PRELAB Value (82 dB))
M2: wp1 dominant pole (794.328 Hz)
V1 (Vertical Marker): Phase Margin: 87 degrees, UGB: 8.49 MHz
GBW: pow (10,80.4567/20) *794.328 = 8.36 MHz (value very close to UGB due to single-poll roll-off)




**Power consumption**: 5*155.2u = 776 uW (higher than what was mentioned in prelab 600uW, primarily due to addition of extra bias branches)

**Key Issues:**

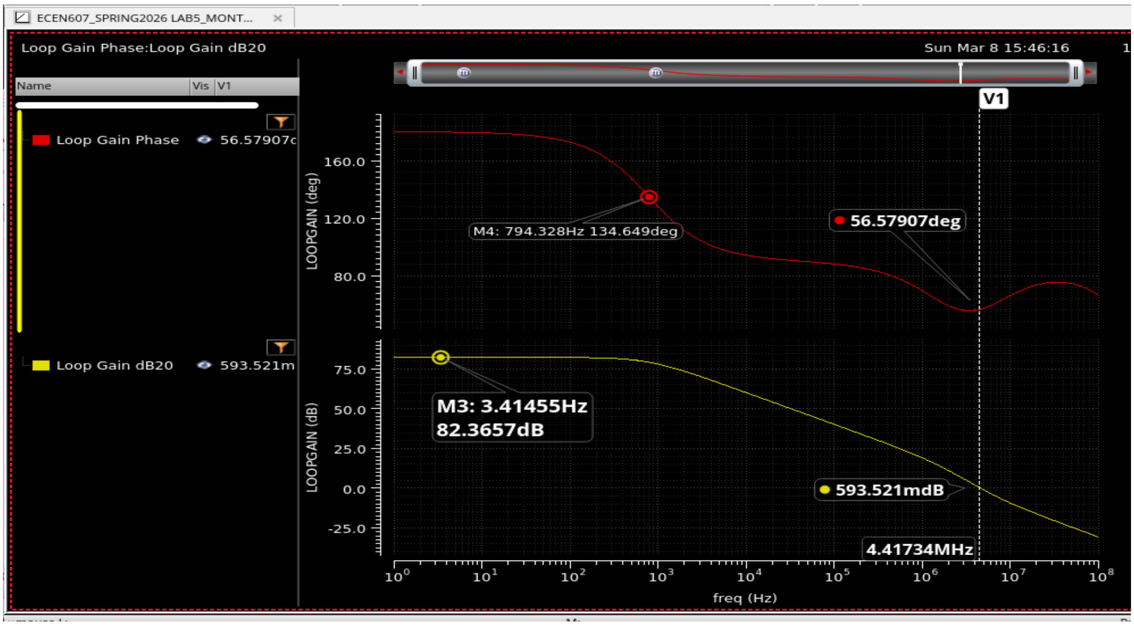
It turns out from the simulation that input-pair gm is not as expected.
Prelab gm1 ~ 175uA/V
Simulation gm1 ~ 114.2 uA/V

Due to which we see a gain reduction of about 2dB and reduction in GBW.
To fix this: input pair sizes were increased to **8*(3/0.5) from 4*(3/0.5).**

**This also increases the GBW without impacting phase margin as non-dominant pole is cancelled by the nulling resistor zero.**

**Another issue** is to save the power, less current was burnt everywhere including the output class-AB branch which means gm was very low which gives rise to very high value of the nulling resistor (Rz = 1/gm *(Cl/Cc +1)) to properly cancel the wp2.
The value of Rz used here is 55Kohms which is (two 110Kohm resistor added in parallel) which is **Area vs Power Trade-off.** Now from the spec-sheet PM required is 45 degrees, so **we are not required to properly cancel the wp2 with zero. That's why value of Rz was reduced from 55Kohms to 17.5Kohms (35Kohms two resistor in parallel).**

**Re-simulating the design with changes:**



**M3: DC Gain (82.36 dB)**
**M4: wp1 (794.328 Hz)**
**V1: Phase margin: 56 degrees (zero was pushed outside and wp2 is at its place)**
**GBW: pow (10,82.36/20) *(794.328) = 10.4 MHz**

**Power consumption: 776 uW**

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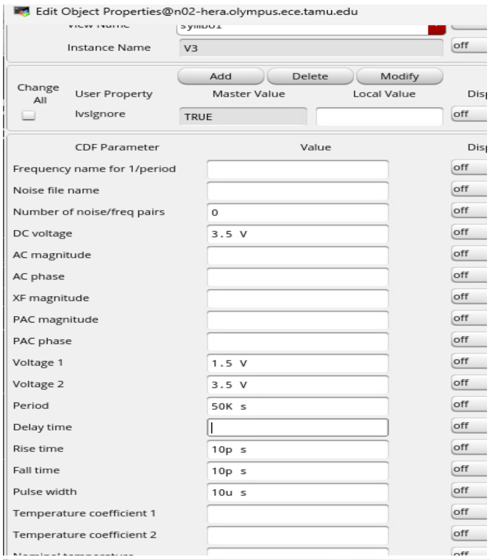
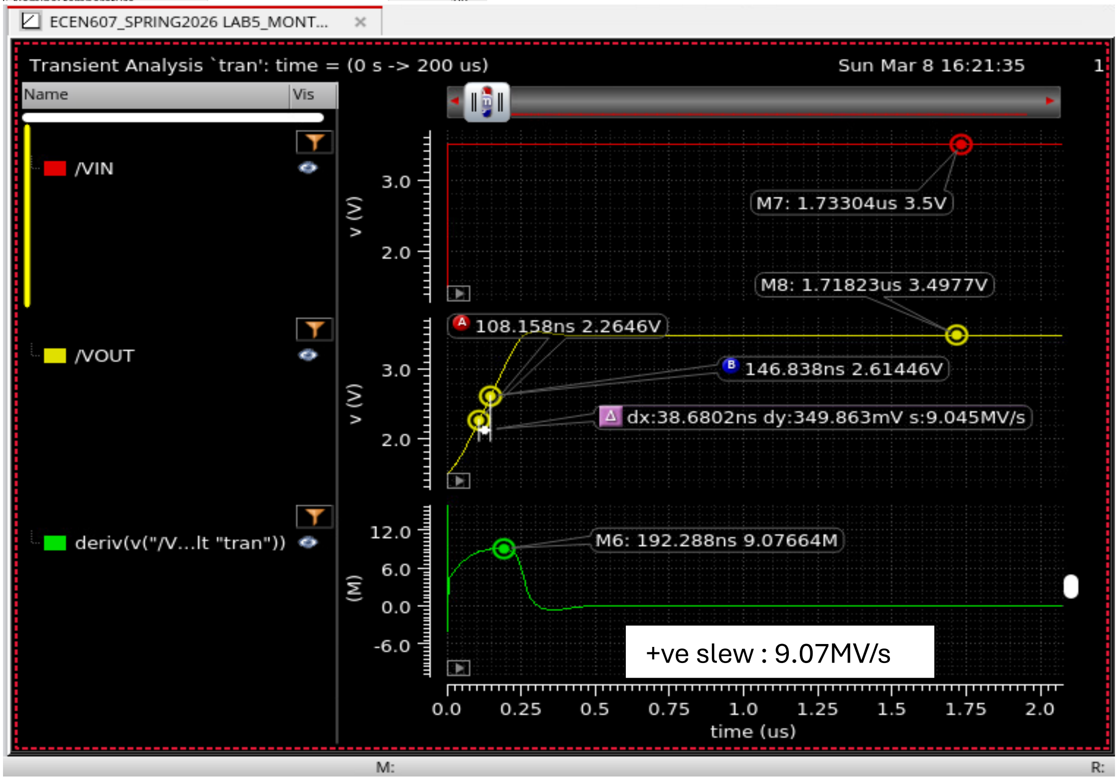
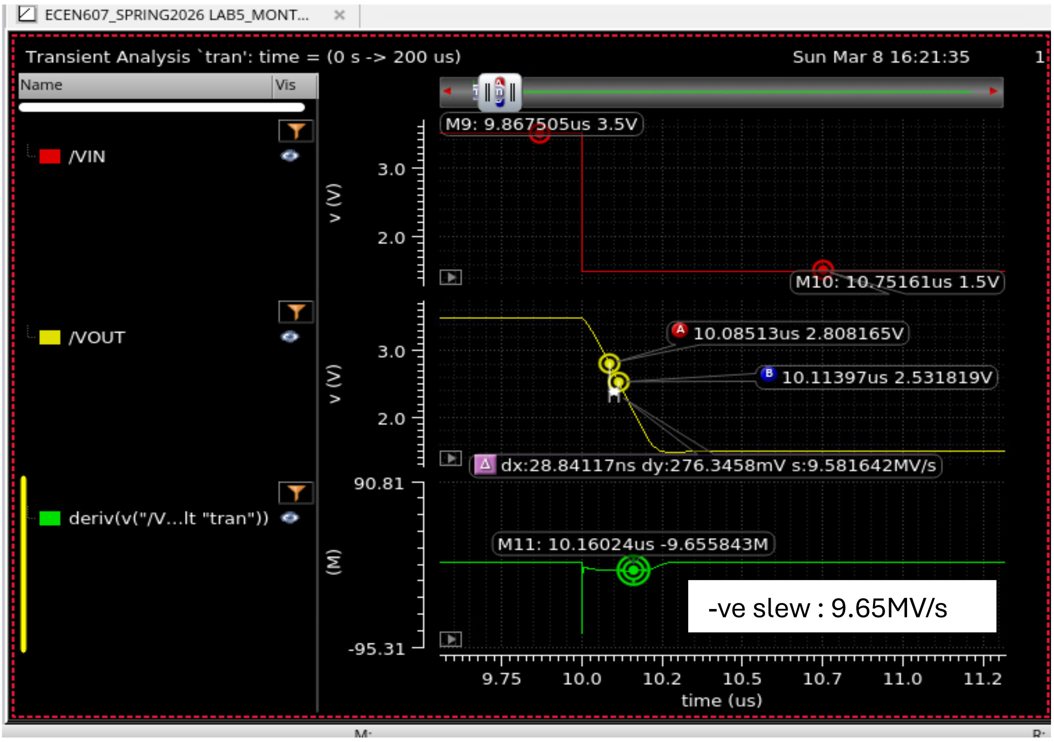
**Slew-Rate Testing:** same closed loop testbench, just the input excitation was changed. In this test it was input pulse with levels between 3.5 and 1.5 Volts. Rise and fall times being higher than the estimated slew rate to properly capture the slew response.





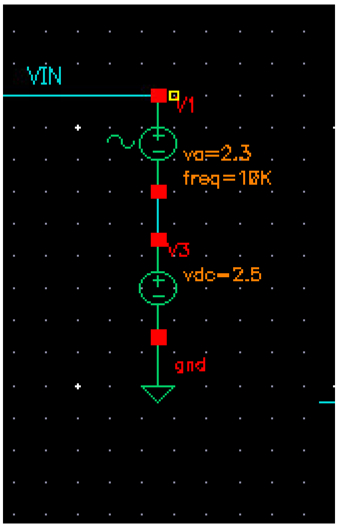

**Output swing testing:**

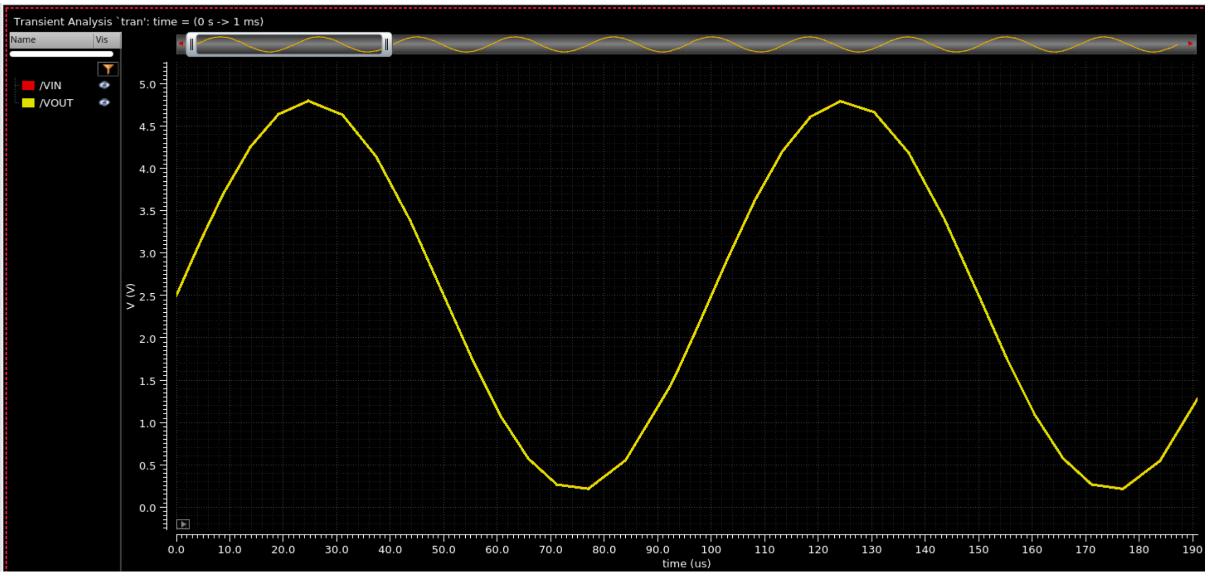
**Please note:** In the design mentioned above all the ideal current sources were replaced by the mos-based current sources. Due to which in the input ICMR is limited on the upper side to provide appropriate headroom to the tail-current source. But on the lower side it can go as low as VSS (**property of folded-cascode which makes them adequate choice for rail-to-rail opamps**).

To test swing: the tail-current source was replaced with ideal-current source which will ensure the input icmr is rail-to-rail and output-swing can be characterized in **closed-loop configuration.**

Stimulus: sine wave (amplitude: 2.3, frequency: 10KHz) on the top of Vdc=2.5 Volts.

Signal is oscillating between 4.8 Volts and 0.2 Volts.






No visible clipping observed, which means swing is sufficiently higher than 4.5 Volts.

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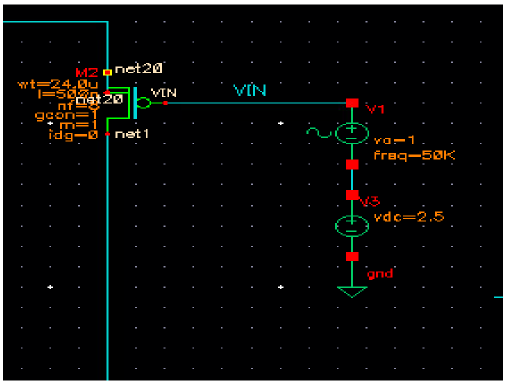
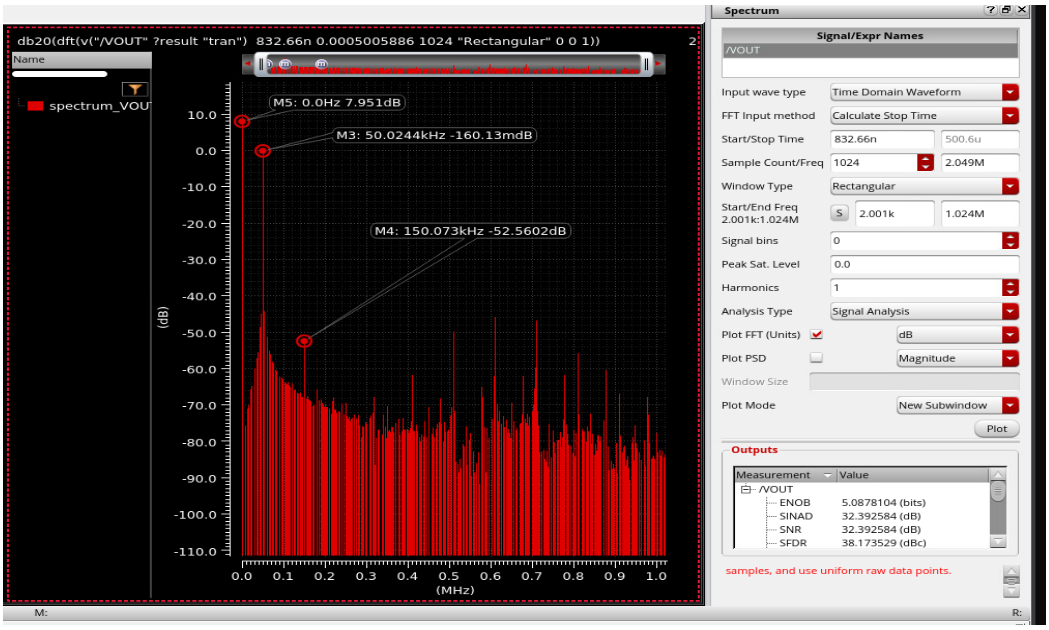
a. THD analysis:
Testbench: Closed-loop setup, Stimulus: Vdc + Vsin = (2.5 + 1.sin(2*pi*50kHz*t))
Sine wave common mode is at 2.5 Volts.

For this test → ideal tail current source was reverted to PMOS based current source.




It will be a transient simulation and output waveform is used to plot the spectrum and calculate the 3rd harmonic.

Fs & Fin were kept co-prime.
1024-point FFT was ran on the output waveform, Rectangular window type.

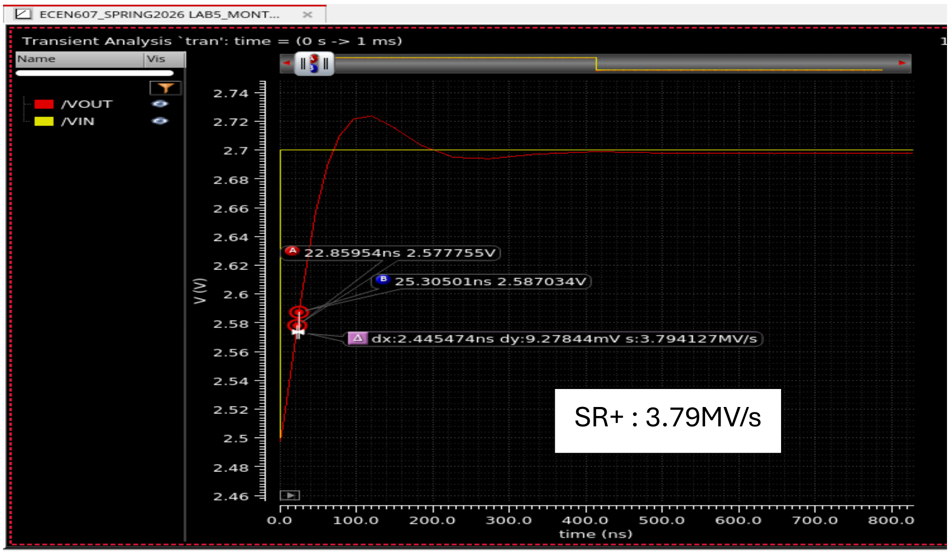



DC component: 2.5 Volts (20log (2.5) = 7.95 dB) → Marker M5
Fundamental Component: Marker M3, -160 mdB
3rd Harmonic component (150kHz): -52.56 dB

Delta_dB (between A1 & A3) = -52.56+0.160 = -52.40 dB
A3/A1 = pow (10, -52.40/20) = 2.4 * 0.001
A3/A1 = 0.24 %

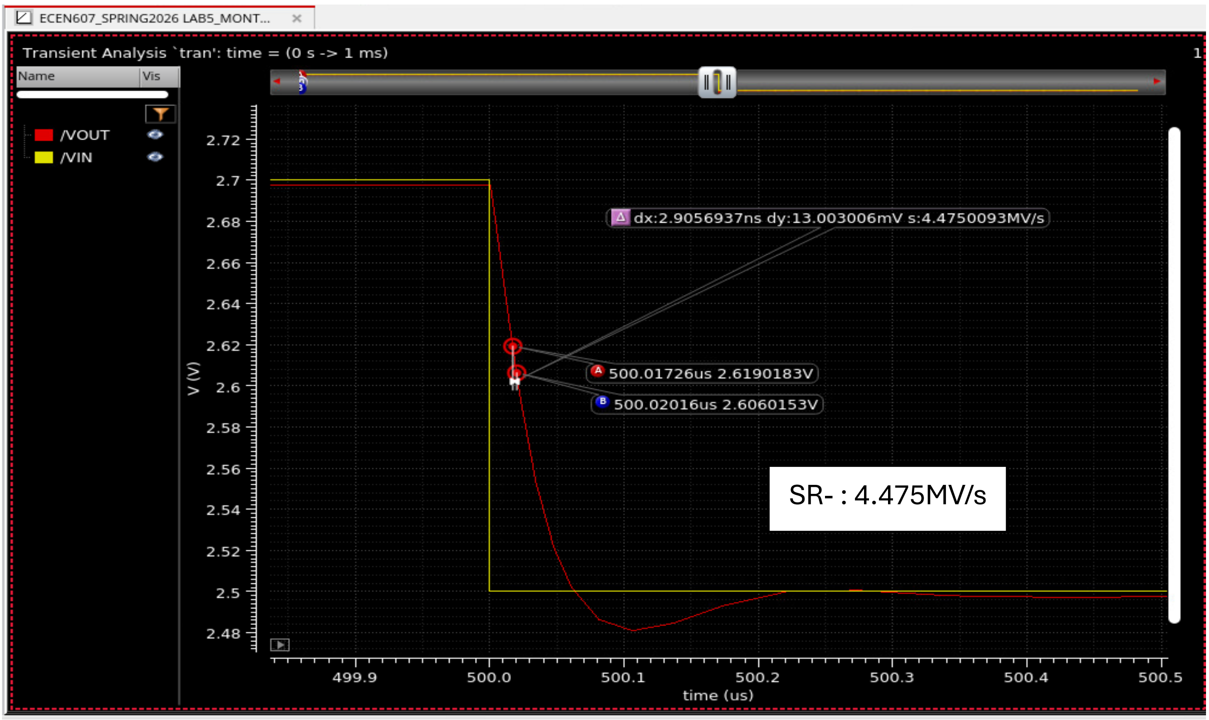
3rd Harmonic ~ 0.24 % of the fundamental (< 1% of the fundamental component)
HD3 = -52.56 + 0.16 = -52.4 dBc ~ 0.24 %

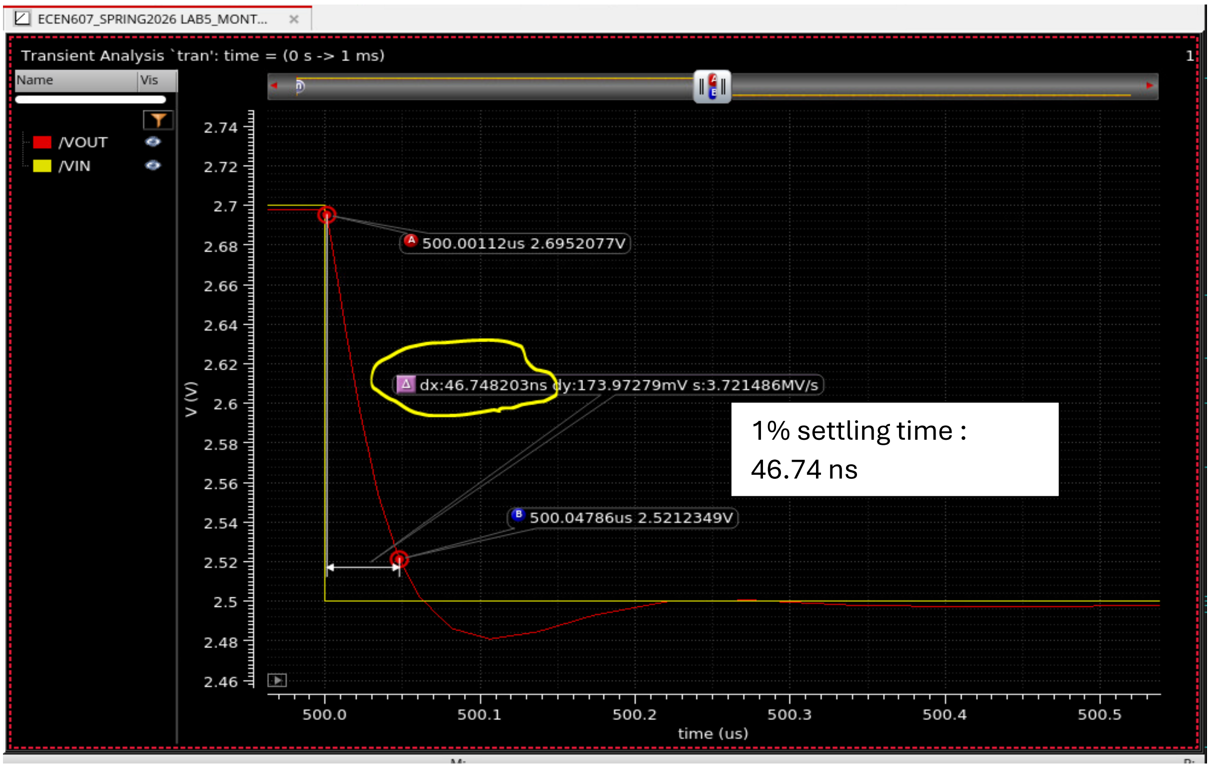
b. Done above
c. Done above
d. Pulse response (Slew rate also measured above)
   Small signal slew rate – 200mV step Frequency 1kHz, settling time measurement:
   Testbench: Closed-loop setup



Please Note: **Small signal slew rate** is primarily defined by the gm of the relevant transistors, so this number is lesser than the large-signal slew rate which was obtained previously.

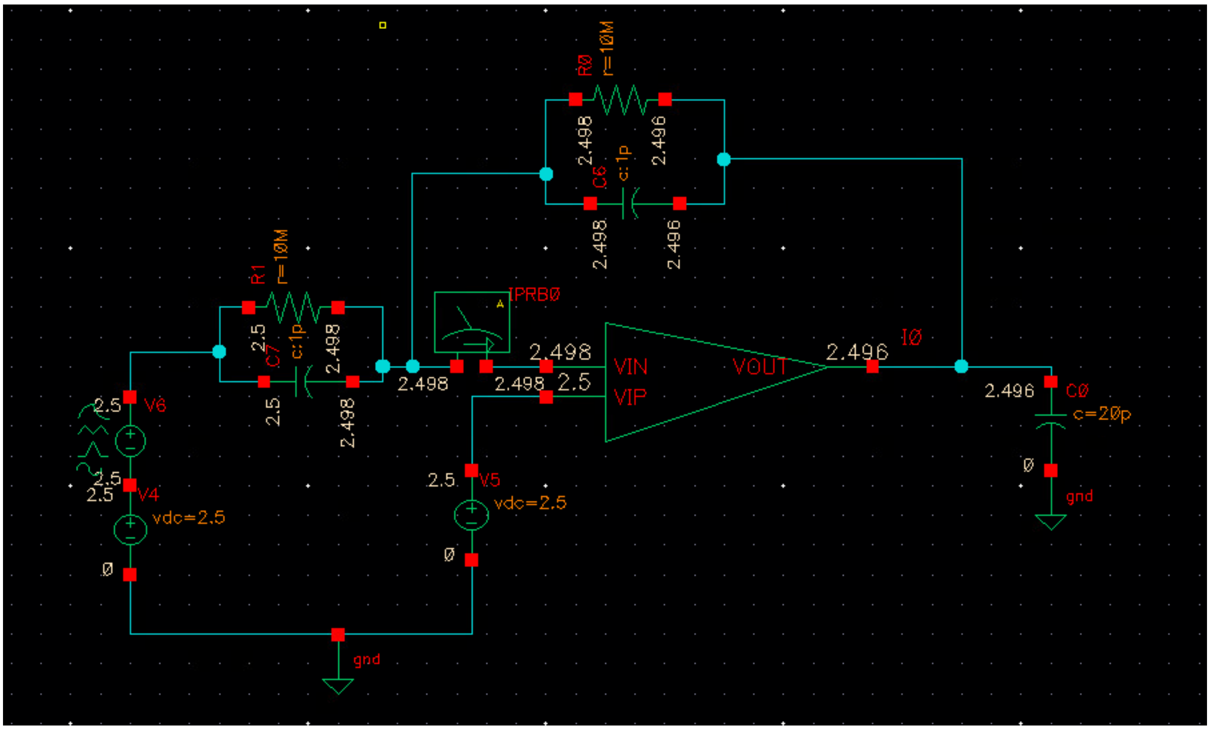
Some rings can also be seen in the output settling due to Phase margin of around 56 degrees.



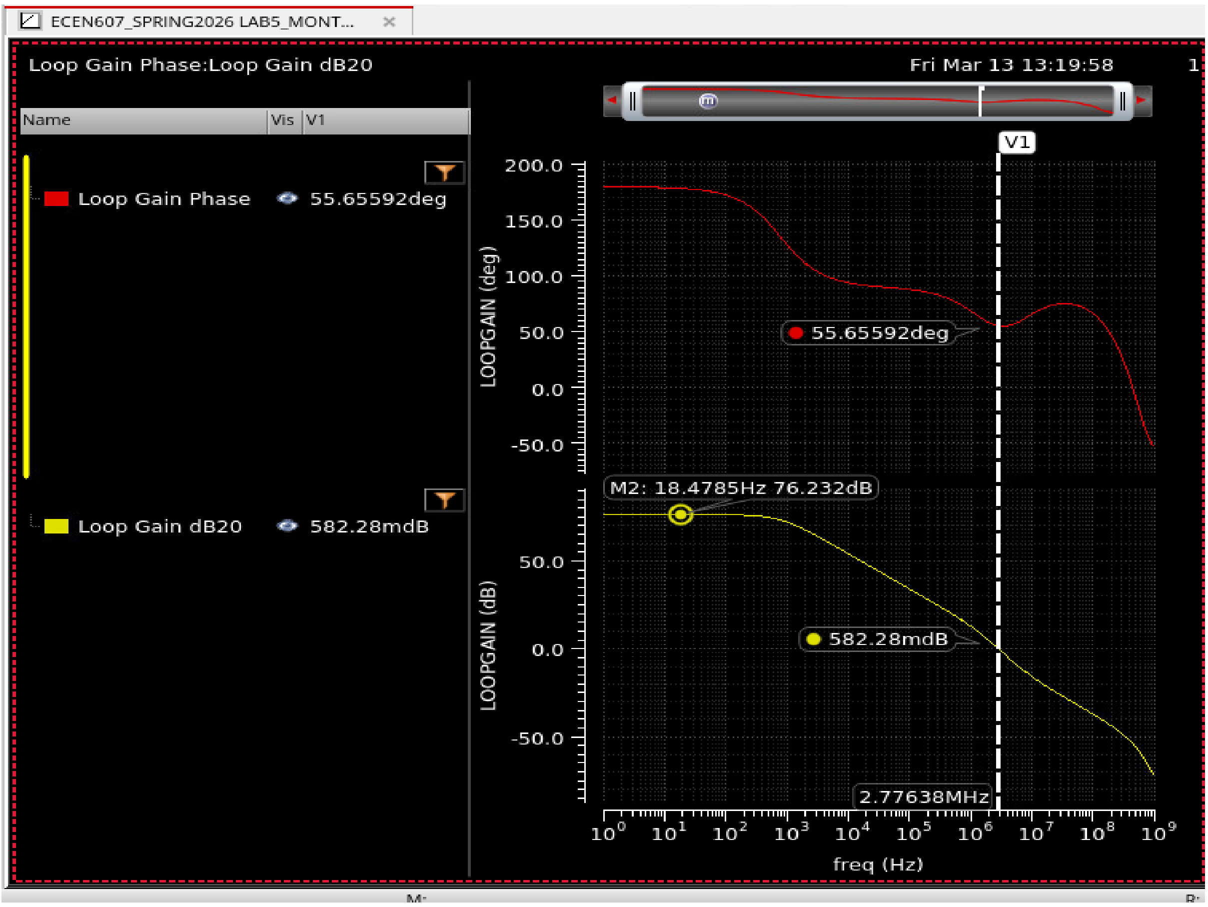

2. Capacitive amplifier, C1, C2 = 1pF, inverting in nature.
   R across C = 10 Mohms (so that it don't load the output of the amplifier)

   Testbench:

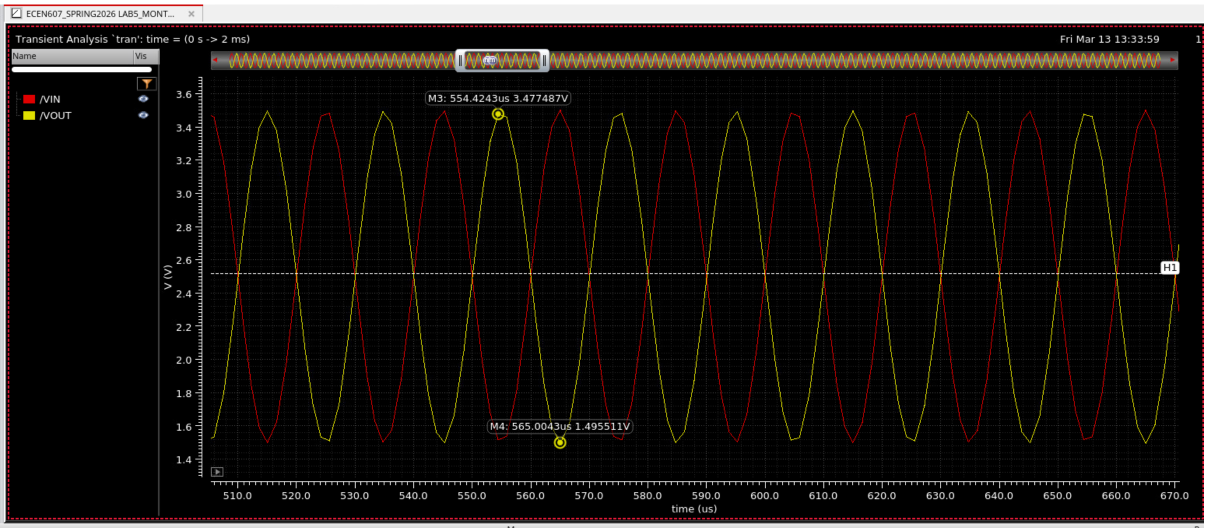



Supply voltages and bias currents are defined within the schematic itself, pins not brought out on the symbol.

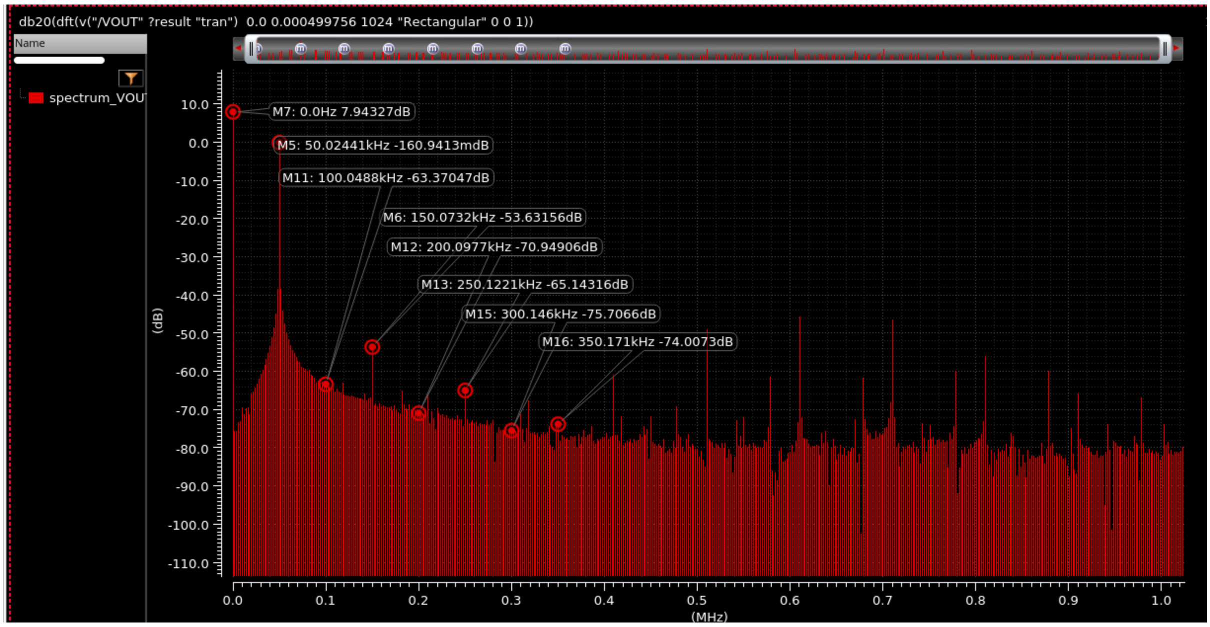
**Sanity-check**: stb run, beta=1/2 so the loop gain should fall by 6dB & ugb should also fall by 2. Can be seen below (Marker M2 gain = 76 dB, opamp gain was around 82 dB)

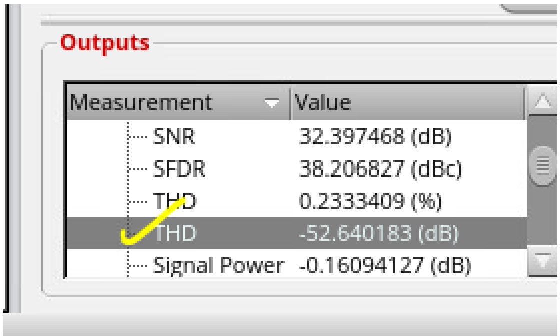



a. Single tone 2Vpp 50kHz.
   Vout & Vin waveform: 180-degree phase shift, wave oscillating between 3.5 Volts and 1.5 volts.



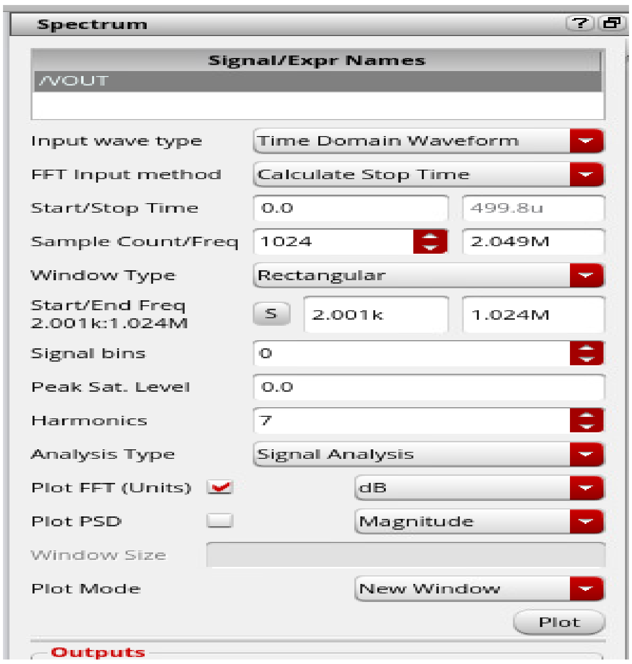

   Spectrum of Vout:
   Fin: 50kHz
   Fs: 2.049 MHz
   Fs & Fin are co-prime.




THD = -52 dB (close to number mentioned in part-1), THD is primarily dominated by 3rd harmonic

FFT settings used:



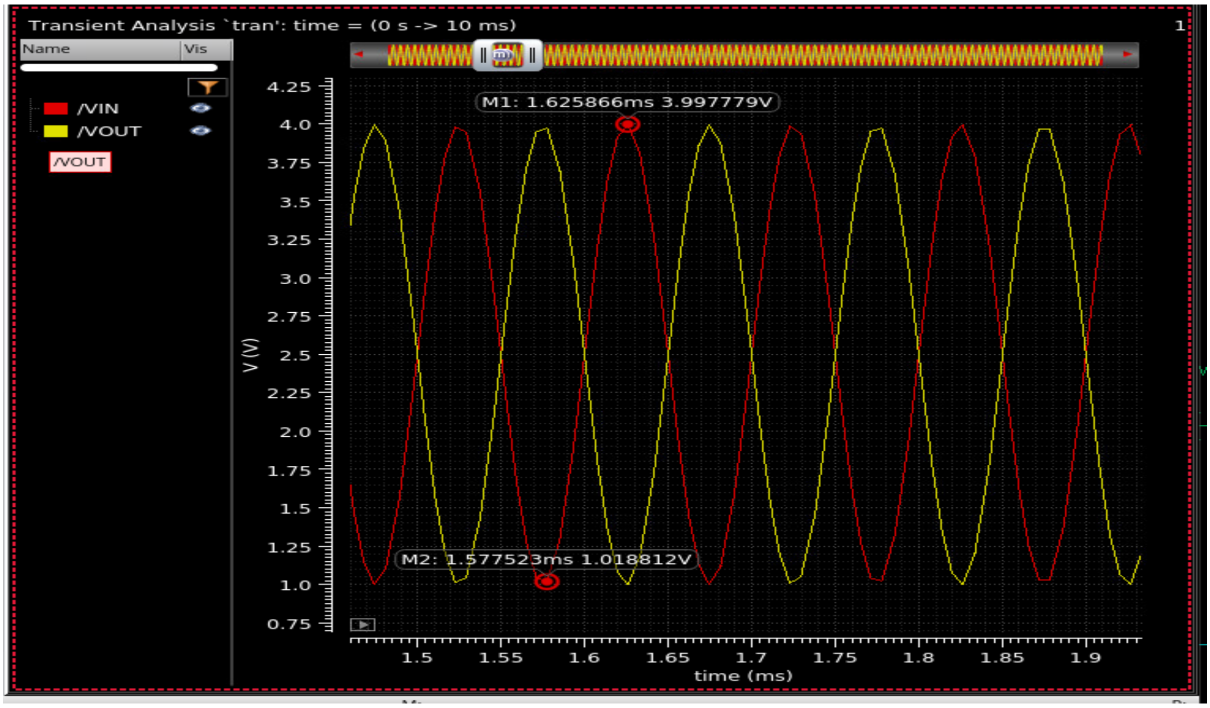
b. Input amplitude sweeps from 200mVpp to 2Vpp

| Point | Test | Output | Nominal |
|---|---|---|---|
| Parameters: Ap_p=200m | | | |
| 1 | ECE... | HD3 | -88.48 |
| 1 | ECE... | HD2 | -83.55 |
| Parameters: Ap_p=400m | | | |
| 2 | ECE... | HD3 | -81.91 |
| 2 | ECE... | HD2 | -77.23 |
| Parameters: Ap_p=600m | | | |
| 3 | ECE... | HD3 | -78.35 |
| 3 | ECE... | HD2 | -73.19 |
| Parameters: Ap_p=800m | | | |
| 4 | ECE... | HD3 | -75.44 |
| 4 | ECE... | HD2 | -70.72 |
| Parameters: Ap_p=1 | | | |
| 5 | ECE... | HD3 | -73.92 |
| 5 | ECE... | HD2 | -69.25 |
| Parameters: Ap_p=1.2 | | | |
| 6 | ECE... | HD3 | -69.14 |
| 6 | ECE... | HD2 | -67.68 |
| Parameters: Ap_p=1.4 | | | |
| 7 | ECE... | HD3 | -62.17 |
| 7 | ECE... | HD2 | -66.66 |
| Parameters: Ap_p=1.6 | | | |
| 8 | ECE... | HD3 | -58.27 |
| 8 | ECE... | HD2 | -69.05 |
| Parameters: Ap_p=1.8 | | | |
| 9 | ECE... | HD3 | -55.89 |
| 9 | ECE... | HD2 | -63.8 |
| Parameters: Ap_p=2 | | | |
| 10 | ECE... | HD3 | -54.12 |
| 10 | ECE... | HD2 | -63.37 |

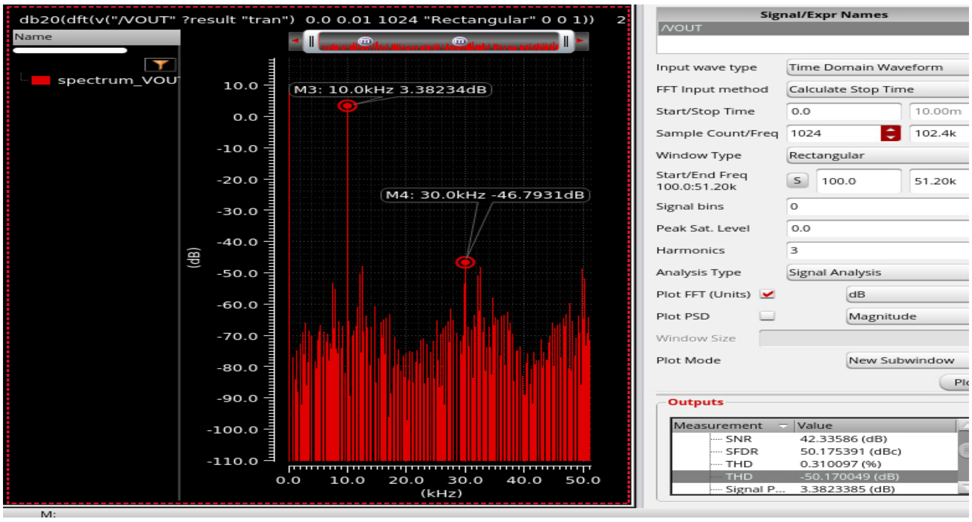
| Name | Type | Details |
|---|---|---|
| HD3 | expr | value(db20(dft(v("/VOUT" ?result "tran") 0.0 0.000499756 1024 "Rectangular" 0 0 1)) 150000) |
| HD2 | expr | value(db20(dft(v("/VOUT" ?result "tran") 0.0 0.000499756 1024 "Rectangular" 0 0 1)) 100000) |

c. HD3 across input frequency (10kHz to 10MHz), while amplitude of the input signal is 3V peak-to-peak.

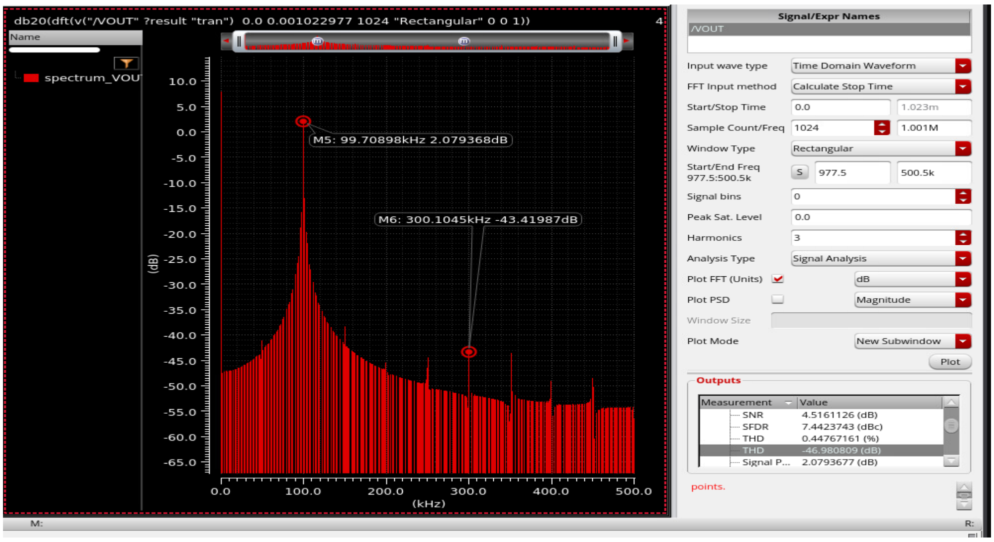
Sims were run for 10kHz, 100kHz, 1MHz, 10MHz (4 points in total)



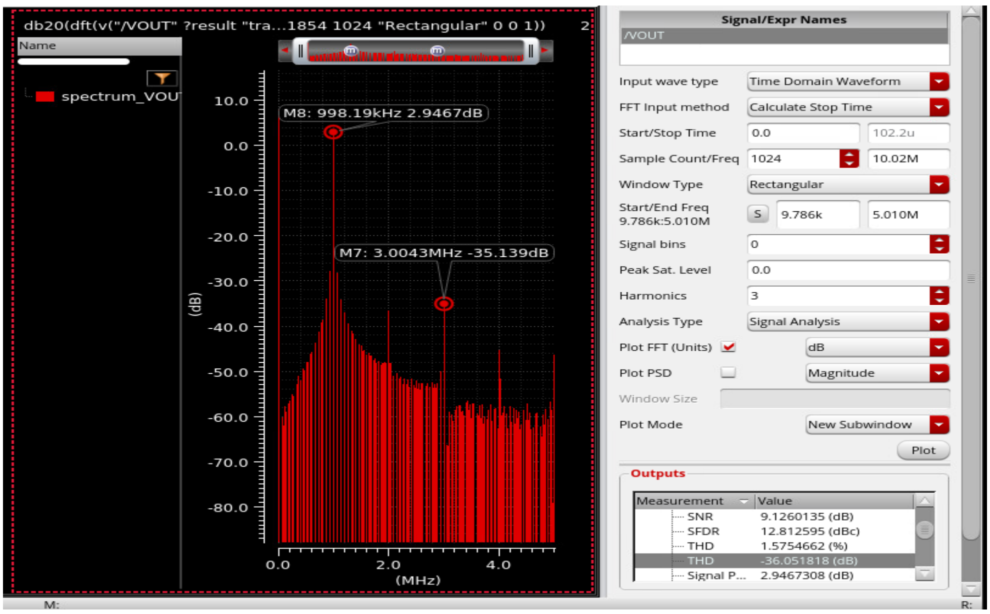
10kHz output spectrum:




100kHz:

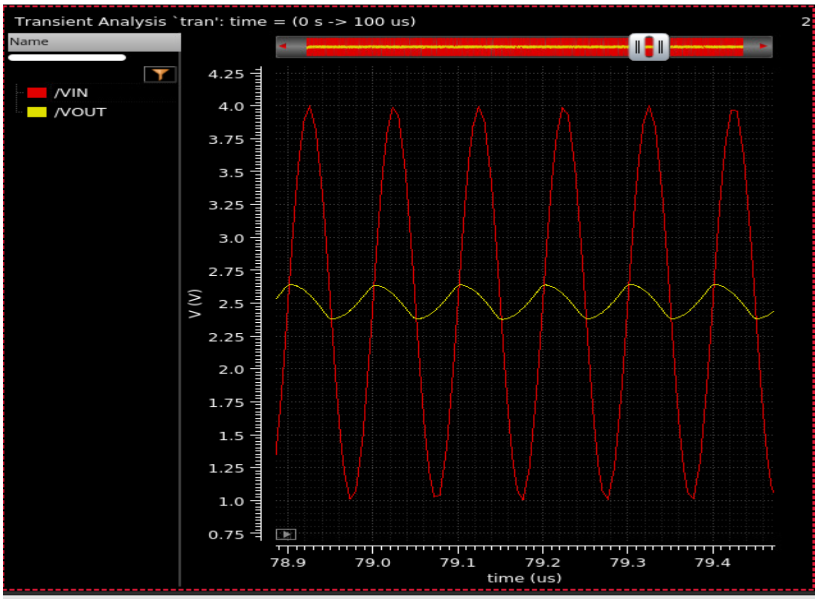


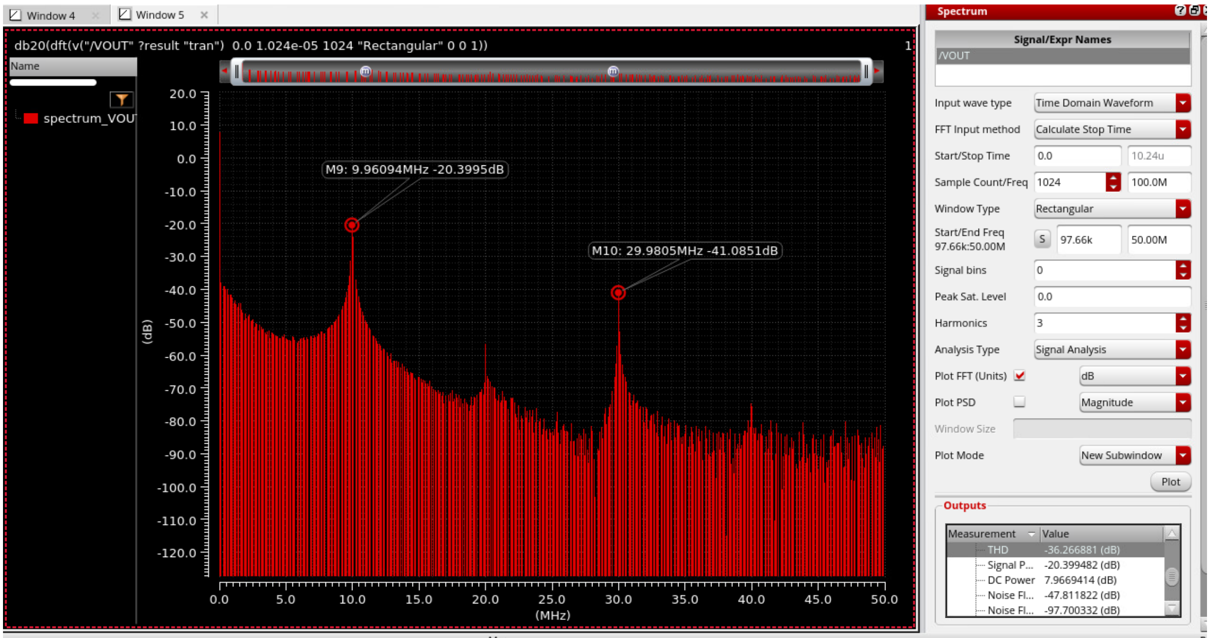

1MHz:



10MHz:

Please Note: This frequency is outside ugb which means loop gain very negligible, so output won't be able to follow the input excitation.




d. Added. Additional Comments: At higher frequency since the gain start to fall, we can see that THD start to worsen. Since the feedback loop start to become weaker.

e. Major sources of non-linearities, ways to improve linearity & trade-offs.
   Major source: Input differential pair, current difference between two input devices becomes nonlinear for large differential input.
   For small signals → linear
   For large signals → HD2, HD3 starts to appear
   Nonlinearity due to current mirror → current mirror non-linearity can be reduced by having transistors with larger length in the circuit to reduce the channel length modulation in the circuit.
   Feedback suppresses lot of this non-linearity.

3. Comparison between prelab & post lab

| PMOS Transistors | W/L_PRELAB | W/L_POSTLAB |
|---|---|---|
| M1,M2 | 4*(3/0.5) | 8*(3/0.5) |
| M3,4,5,6,12,13 | 8*(0.5/0.5) | 8*(0.5/0.5) |
| M11a | 4*(0.5/0.5) | 4*(0.5/0.5) |
| M16 | 8*(0.5/0.5) | 8*(0.5/0.5) |
| M18 | 2*(0.5/0.8) | 2*(0.5/0.6) |

| NMOS transistors | W/L_PRELAB | W/L_POSTLAB |
|---|---|---|
| M7,9,14,15 | 4*(0.7/0.7) | 4*(0.7/0.7) |
| M8,10 | 8*(0.7/0.7) | 8*(0.7/0.7) |
| M11b | 2*(0.7/0.7) | 2*(0.7/0.7) |
| M17 | 4*(0.7/0.7) | 4*(0.7/0.7) |
| M19 | 2*(0.7/1) | 2*(0.7/1) |
| M20,21 | 8*(0.7/0.7) | 8*(0.7/0.7) |

| | PRELAB | POSTLAB |
|---|---|---|
| Rz | 55Kohms | 35Kohms |
| Cc | 2pF | 2pF |

| Parameters | SPECS | PRELAB | Simulations |
|---|---|---|---|
| Av0 | >60dB | 82dB | 82.36dB |
| Slew Rate | 4MV/s | | 9MV/s |
| GBW | >10MHz | 13.9MHz | 10.4MHz |
| PM | >45 | 90 | 56 |
| Output Swing | >4.5 | 4.6 | >4.5 |
| CL | 20pF | | |
| Power | <1mW | 0.6mW | 0.776mW |

4. 1dB compression pt.

This is the signal amplitude at which gain of the amplifier falls by 1-dB.

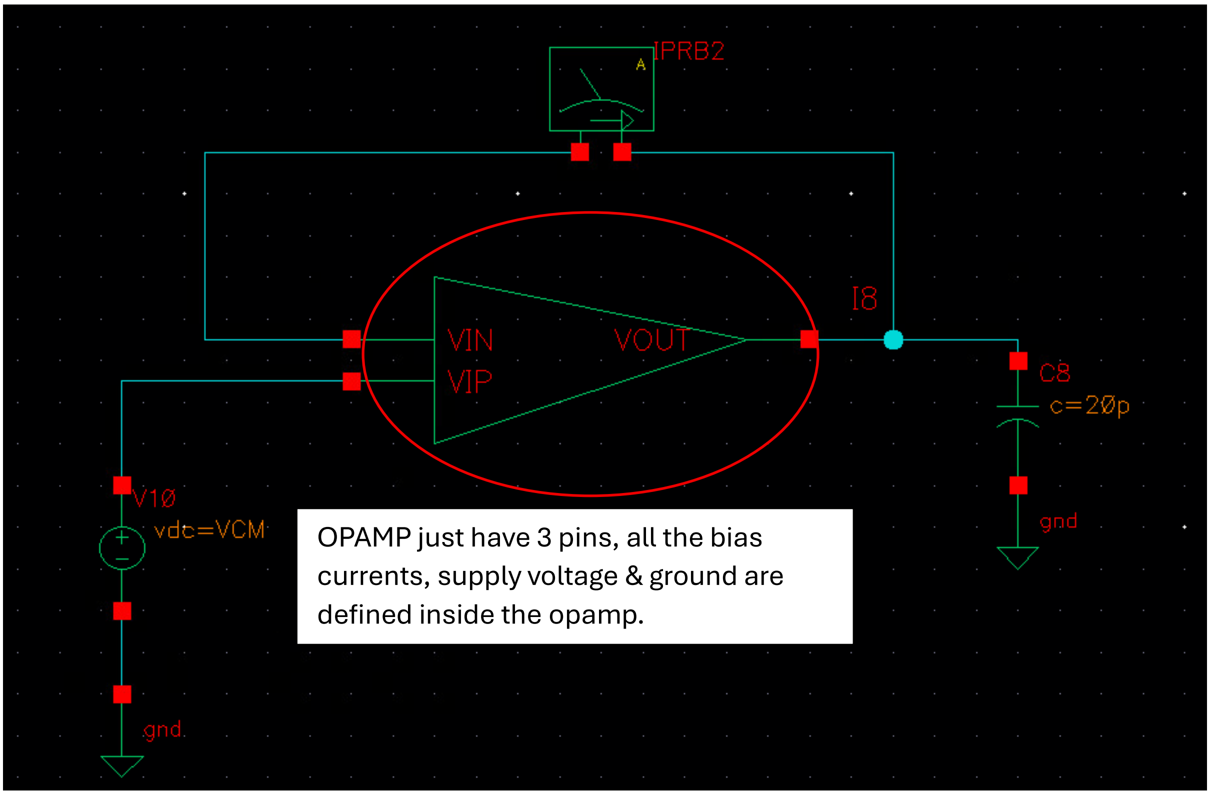
Now,

- ➔ We can't test this amplifier in open-loop topology since the output bias point is not very well-defined. (Ideally it should be 2.5 Volts, mid-rail but comparison between nmos and pmos current sources is not always in control so the output can be seen to be skewed in one direction).
- ➔ Now we can't run PSS sim with closed-loop opamp topology since it will be giving us 1-dB compression pt for the opamp in closed-loop (which have a gain of 1 if connected in unity-gain configuration), we want where just the gain of opamp falls by 1-dB (in my case the gain at proper bias point was around 82dB, so need to find signal amplitude where the gain falls to 81dB)

**Method Used:**

OPAMP was connected in closed-loop (unity-gain buffer configuration). Probe was connected and "stb" analysis was ran across VCM of 0.2 Volts to 4.8 Volts. Low-Frequency gain was monitored (this opamp is more of a baseband opamp).

TESTBENCH:




Results: **Low-frequency Gain vs input CM**


Assuming M2 to be optimal bias point (mid-rail)
M5: represents the point where gain falls by 1dB from the gain at optimal bias point.

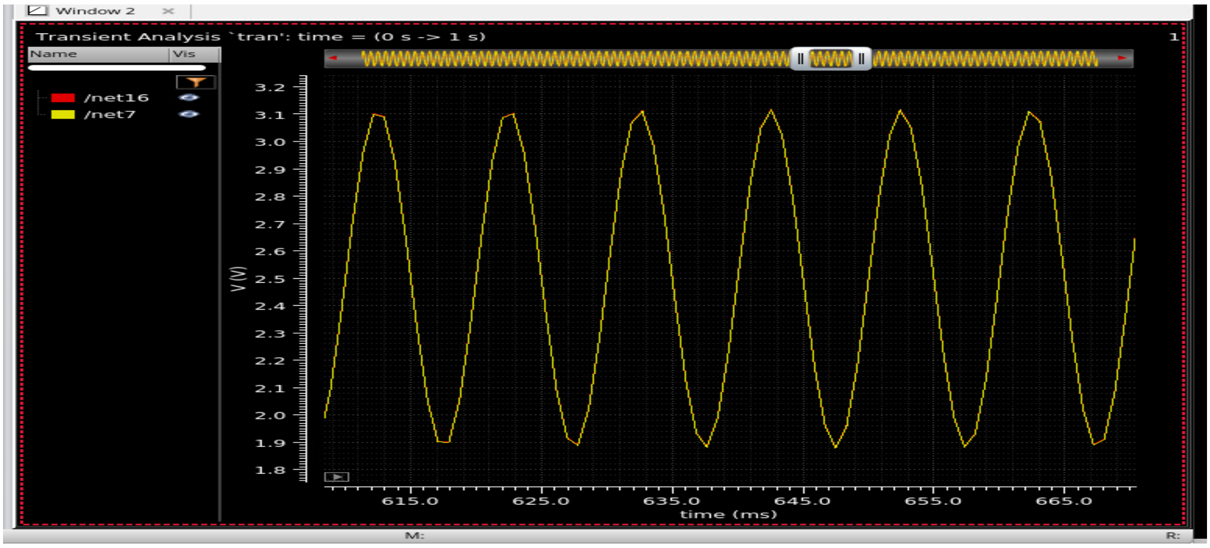
1-dB compression point @ Rx side represents the input signal swing required to reduce the gain by 1-dB.
which means if there is a sinusoid of Amplitude = 3.18-2.5 = 0.618 Volts

From RFIC it's known that **0 dBm → 632mVpp (50-ohm resistor) / 316mV-peak**

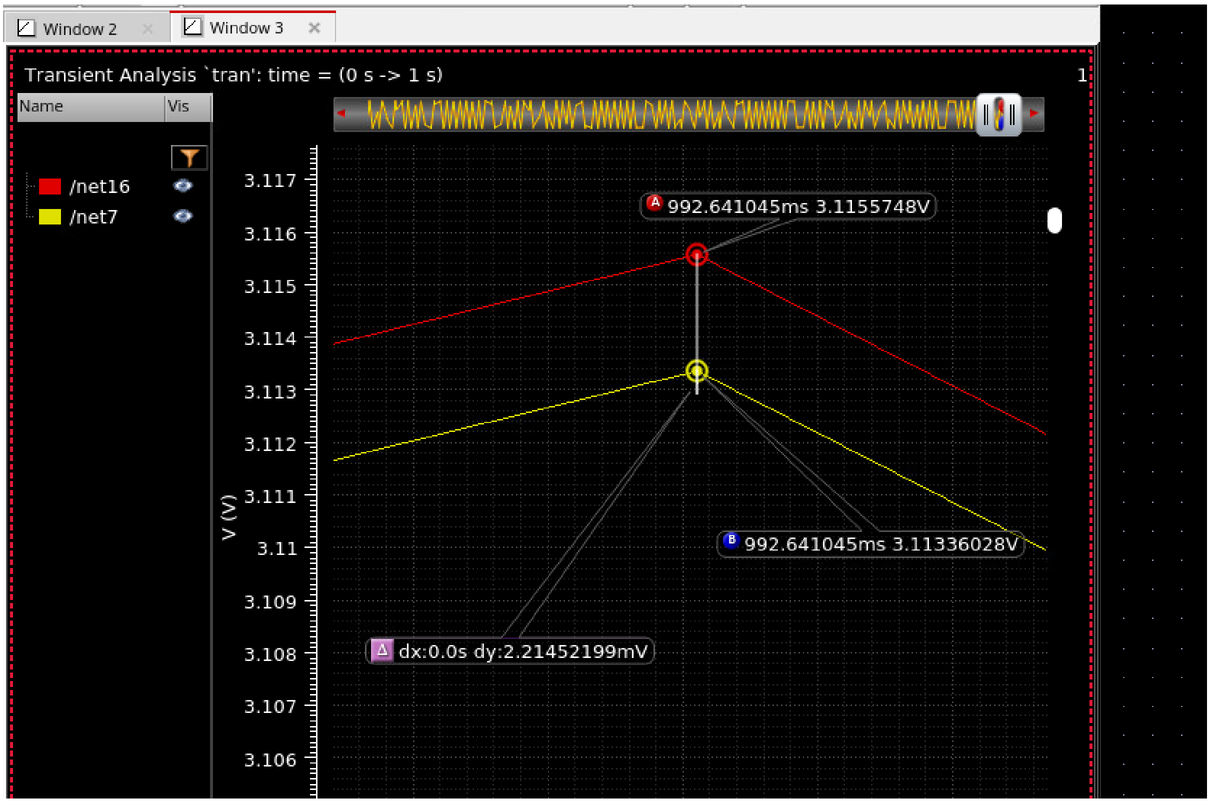
So, **P-1dB = 618mV-peak ~ 5.82 dBm**

**Transient-Approach: VCM: 2.5 Volts, Sinusoid of A=0.618 Volts, F=100 Hz was applied at the input and output was monitored.**

**/Net7 – output , /Net16 – input**




Zoomed View around the peak to examine the gain-error:




A = Vout/(Vout-Vin) = 3.11336028/ (2.21452199E-3) = 1405 ~ 63 dB

This don't represent the real gain since there is some input referred systematic offset error as well apart from "gain error". So, calculating gain from the difference between Vout & Vin might be the right approach in this type of cases.

5. Final Comments & Conclusions:

- ➔ Learnt Class AB biasing for fully rail-to-rail output.
- ➔ Learnt Area vs Power trade-off → By having higher Rz compensation resistor power can be saved as lesser current is burnt in the output stage to make the system stable.
- ➔ Understanding of harmonic distortion
- ➔ Slew-Rate understanding → Even after having a output stage which can supply/sink lot of current, our slew was limited primarily due to first stage bias-current and cap at the output of first stage.
- ➔ From results achieved point of view: All the specs were met with enough margin, maybe to increase GBW some different compensation schemes could have been used like **feedforward compensation or cascode compensation schemes.**